$= -8,00,000[0.3396 - 1]$

$= 5,28,320$

Cost of the machine $= 5,\ 00,\ 000 - 30,\ 000$

$= 4,\ 70,\ 000$

Hence it not advisable to rent the machine

It is better to buy the machine.

Inventory :

Given the inventory on hand $I(x)$ and the unit holding cost (C_1), the total inventory carrying cost is $C_1 \int_0^T I(x)dx$, where T is the time period under consideration.

Example 3.22

A company receives a shipment of 200 cars every 30 days. From experience it is known that the inventory on hand is related to the number of days. Since the last shipment, $I(x) = 200 - 0.2x$. Find the daily holding cost for maintaining inventory for 30 days if the daily holding cost is ₹3.5.

Solution:

Here $I(x) = 200 - 0.2x$

$C_1 = ₹\ 3.5$

$T = 30$

Total inventory carrying cost

$$= C_1 \int_0^r I(x)dx = 3.5 \int_0^{30} (200 - 0.2x)dx$$

$$= 3.5\left(200x - \frac{0.2x^2}{2}\right)_0^{30} = 20,685$$

Amount of an Annuity

The amount of an annuity is the sum of all payments made plus all interest accumulated. Let an annuity consist of equal payments of Rs. p and let the interest rate of r percent annually be compounded continuously.

Amount of annuity after N payments

$$A = \int_0^N p e^{rt}\, dt\,.$$

Example 3.23

Mr. Arul invests ₹10,000 in ABC Bank each year, which pays an interest of 10% per annum compounded continuously for 5 years. How much amount will there be after 5 years. $\left(e^{0.5} = 1.6487\right)$

Solution:

$p = 10000,\quad r = 0.1,\quad N = 5$

Annuity $= \int_0^5 10000\, e^{0.1t}\, dt$

$$= \frac{10000}{0.1}\left(e^{0.1t}\right)_0^5$$

$$= 100000\left[e^{0.1\times 5} - e^0\right]$$

$$= 100000\left(e^{0.5} - 1\right)$$

$$= 100000\left[0.6487\right]$$

$= ₹\ 64,870$

Consumption of a Natural Resource

Suppose that $p(t)$ is the annual consumption of a natural resource in year t. If the consumption of the resource is growing exponentially at growth rate k, then the total consumption of the resource after T years is given by

$$\int_0^T p_0\, e^{kt}\, dt = \frac{p_0}{k}\left(e^{kT} - 1\right)$$

Where p_0 is the initial annual consumption at time $t = 0$.

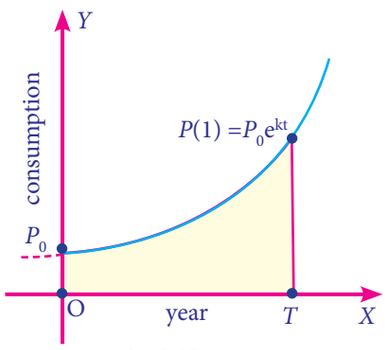


Fig. 3.15

Example 3.24

In year 2000 world gold production was 2547 metric tons and it was growing

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